

What is a Determinant

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Abstract

We derive the determinant of an $n \times n$ matrix from three axioms motivated by oriented volume, prove $|A| = |A^T|$ by reindexing the permutation sum, then compute the derivative of the determinant (*Jacobi's formula*) and apply it to deduce *Liouville's theorem* on the volume evolution of linear flows $\dot{\mathbf{x}} = A\mathbf{x}$.

1 From oriented volume to the determinant formula

Most of us meet the determinant as a formula – $a_{1,1}a_{2,2} - a_{1,2}a_{2,1}$ for a 2×2 matrix, with a cofactor recipe for anything larger – together with instructions to compute it, but with no hint of where the formula comes from or what it is *for*. This paper takes the opposite route: we start from a question one might actually ask, and let the formula be forced.

Here is the question. When an $n \times n$ matrix A acts on R^n , every region of space is stretched, sheared, or flipped. *By what factor does volume change?* This one number is worth wanting for at least two reasons: it is exactly the factor that the change-of-variables formula of multivariable calculus must insert; and – as will fall out of the theory almost for free – a matrix that flattens volume to zero cannot be inverted.

The answer will turn out to be the *signed* factor by which the volume of the unit cube is changed upon the action of A – the *determinant* of A . (We will see that the sign is forced by the algebra; for now treat it as a feature, not a bug.) The cube is transformed into a parallelepiped. The n edges of the unit cube meeting at the origin are the standard basis vectors: $\{[1, 0, \dots, 0], [0, 1, 0, \dots, 0], \dots, [0, \dots, 0, 1]\}$. When A acts on these vectors through matrix multiplication it produces the column vectors of A – by the definition of matrix multiplication.¹

So, to find the determinant – the stretching factor of A – we need to compute the volume of the parallelepiped formed from the column vectors of A . This is enough as the stretching factor is the ratio of the volume after applying A to the volume of what we started with. Since the starting point is the unit cube which has a volume of 1, the ratio *is* the volume of the stretched unit cube under the action of A .

Given n vectors in R^n , how do we find a formula for the parallelepiped volume? A good way is to use the properties of the function to help find a formula.

¹That $A\mathbf{e}_j$ is the j^{th} column of A is itself forced, not decreed: in the companion paper “Origins of Matrix/Vector and Matrix/Matrix Multiplication” the definition of matrix multiplication is derived from the demand that the scalar solution of $a x = b$ replay for systems of equations.

Let's call our volume function $V(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. Here are some properties we should agree on:

Multi-linear Property: V is linear in each slot. That is, it is a multi-linear function.

Degenerate Property: If any two edges are the same then the volume is 0.

Volume of Canonical Unit Cube: The volume of the unit cube using standard ordering of unit edges, $\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n$, is 1.

Let's go through each property. We are not deriving these from a prior definition of geometric volume; rather, we are *postulating* a function V with these properties, motivated by what oriented volume should look like. Scaling is intuitive: if we scale any slot (edge) by a factor k then the volume of the resulting parallelepiped should scale by k . Additivity in each slot is more subtle and is genuinely a postulate rather than a visible fact about parallelepipeds — but it is the assumption that lets the rest of the theory go through, and the resulting function turns out to coincide with geometric volume in the cases ($n = 2, 3$) where we can check.

The degeneracy property is clear, if we have duplicate edges then a parallelogram in R^2 collapses to a line segment and consequently the area is zero. Similarly for higher dimensions.

The third property is clearly true, but why the need to care about the ordering of the standard edges? We show why by looking at a property implied by the multi-linearity and degeneracy properties.

Let's focus on R^3 as the general case is similar. Notice that

$$V(\mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_3) = 0 \quad (1)$$

because if the same vector is in two slots the volume is 0. But by the multi-linearity (1) becomes:

$$\begin{aligned} 0 &= V(\mathbf{x}_1, \mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_1 + \mathbf{x}_2, \mathbf{x}_3) \\ &= V(\mathbf{x}_1, \mathbf{x}_1, \mathbf{x}_3) + V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_2, \mathbf{x}_3) \\ &= V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) + V(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3) \end{aligned} \quad (2)$$

From this we see that

$$V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = -V(\mathbf{x}_2, \mathbf{x}_1, \mathbf{x}_3) \quad (3)$$

That is, V is *anti-symmetric*.

So, the reason the third property requires the unit cube to be ordered is that order matters because of this anti-symmetry property. *That is, a consequence of our assumptions is that the volume function will be signed!*

Let's now use these two properties to derive a formula for the volume function.

Each of the vectors: $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ can be written in terms of the standard basis: $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ as

$$\begin{aligned} \mathbf{x}_1 &= a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2 + a_{3,1}\mathbf{e}_3 \\ \mathbf{x}_2 &= a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2 + a_{3,2}\mathbf{e}_3 \\ \mathbf{x}_3 &= a_{1,3}\mathbf{e}_1 + a_{2,3}\mathbf{e}_2 + a_{3,3}\mathbf{e}_3 \end{aligned}$$

If $\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3$ are the mapping of $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$ by matrix A , then A is related to the $\{a_{i,j}\}_{i,j=1}^n$ by:

$$A = \begin{bmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{bmatrix} \quad (4)$$

So, coming up with a formula for the volume function in terms of the coefficients $a_{i,j}$ will translate directly to a formula for the determinant of A .

To illustrate the process of finding the volume we will do it for the case when $n = 2$. In this case we have:

$$\begin{aligned} V(\mathbf{x}_1, \mathbf{x}_2) &= V(a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2, a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2) \\ &= V(a_{1,1}\mathbf{e}_1, a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2) + V(a_{2,1}\mathbf{e}_2, a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2) \\ &= V(a_{1,1}\mathbf{e}_1, a_{1,2}\mathbf{e}_1) + V(a_{1,1}\mathbf{e}_1, a_{2,2}\mathbf{e}_2) \\ &\quad + V(a_{2,1}\mathbf{e}_2, a_{1,2}\mathbf{e}_1) + V(a_{2,1}\mathbf{e}_2, a_{2,2}\mathbf{e}_2) \\ &= a_{1,1}a_{1,2}V(\mathbf{e}_1, \mathbf{e}_1) + a_{1,1}a_{2,2}V(\mathbf{e}_1, \mathbf{e}_2) \\ &\quad + a_{2,1}a_{1,2}V(\mathbf{e}_2, \mathbf{e}_1) + a_{2,1}a_{2,2}V(\mathbf{e}_2, \mathbf{e}_2) \end{aligned}$$

Notice that we could pull out the coefficients $a_{i,j}$ because of multi-linearity – linearity refers not just to addition, but also scalar multiplication. Also, we have shown that the degeneracy condition implies anti-symmetry: $V(\mathbf{e}_1, \mathbf{e}_2) = -V(\mathbf{e}_2, \mathbf{e}_1)$. Finally, we know the volume (area in this case) of $V(\mathbf{e}_1, \mathbf{e}_2)$. It represents the volume of the unit cube (unit square in our case) and its volume is 1. Therefore,

$$V(\mathbf{x}_1, \mathbf{x}_2) = a_{1,1}a_{2,2} - a_{1,2}a_{2,1} \quad (5)$$

This corresponds with the usual definition of the determinant of the matrix A . Pause on that: this is a *calibration*. The machine built from the three properties reproduces the 2×2 formula the reader already owns – which is how one learns to trust a new tool. It is also the one place we can audit the postulates against honest geometry: a base-times-height computation gives the area of the parallelogram with edges $\mathbf{x}_1, \mathbf{x}_2$ as $|a_{1,1}a_{2,2} - a_{1,2}a_{2,1}|$, so the postulated additivity has not led us away from true area.

In the general case, we will have to do all of the permutations of the indices of a upon re-ordering the base vectors into standard form.

Let's see what this looks like in R^3 .

$$\begin{aligned} V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) &= V(a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2 + a_{3,1}\mathbf{e}_3, \\ &\quad a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2 + a_{3,2}\mathbf{e}_3, \\ &\quad a_{1,3}\mathbf{e}_1 + a_{2,3}\mathbf{e}_2 + a_{3,3}\mathbf{e}_3) \end{aligned} \quad (6)$$

By multi-linearity this can be decomposed into the following:

$$V(\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3) = \sum_{i=1}^3 \sum_{j=1}^3 \sum_{k=1}^3 a_{i,1}a_{j,2}a_{k,3}V(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k) \quad (7)$$

There are 3^3 terms to consider. In the general case, there will be n^n terms to deal with – this is not looking promising. But notice, not all terms, $V(\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k)$ survive. Any such term with a duplicate $\mathbf{e}_i$ will have value

0 because of the degeneracy condition. What terms do survive? The terms where the indices, i, j, k , differ. That is, i, j, k are *permutations* of $1, 2, 3$.² In the general case this reduces the number of terms in the sum from n^n to $n!$. In the general case, the sum becomes:³

$$V(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}) \quad (8)$$

Now, how do we compute each of the terms: $V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)})$? We know that we can switch the values of any two of the n slots and the value of V will change signs. It turns out that for any permutation, one can apply some number of these switches (called *transpositions*) to get back to the canonical ordering $V(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$. So the value of $V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)})$ should be the value of $V(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n)$ times $(-1)^k$, where k is the number of transpositions used to get to the canonical ordering of indices. We denote the sign of σ by $\text{sgn}(\sigma) \in \{-1, +1\}$.⁴ And since we know the volume of the canonical cube is 1, the formula for the parallelepiped – and hence the determinant of the matrix A – becomes:

$$\begin{aligned} |A| &= V(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) \\ &= V(a_{1,1}\mathbf{e}_1 + a_{2,1}\mathbf{e}_2 + \dots + a_{n,1}\mathbf{e}_n, \\ &\quad a_{1,2}\mathbf{e}_1 + a_{2,2}\mathbf{e}_2 + \dots + a_{n,2}\mathbf{e}_n, \\ &\quad \vdots \\ &\quad a_{1,n}\mathbf{e}_1 + a_{2,n}\mathbf{e}_2 + \dots + a_{n,n}\mathbf{e}_n) \quad (\text{expanding vectors into components}) \\ &= \sum_{j_1=1}^n \sum_{j_2=1}^n \dots \sum_{j_n=1}^n a_{j_1,1} a_{j_2,2} \dots a_{j_n,n} V(\mathbf{e}_{j_1}, \mathbf{e}_{j_2}, \dots, \mathbf{e}_{j_n}) \quad (\text{by the Multi-linearity Property}) \\ &= \sum_{\sigma \in S_n} a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} V(\mathbf{e}_{\sigma(1)}, \mathbf{e}_{\sigma(2)}, \dots, \mathbf{e}_{\sigma(n)}) \quad (\text{by the Degeneracy Property}) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} V(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) \quad (\text{after placing edges in Canonical Order}) \\ &= \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} \quad (\text{Volume of Canonical Unit Cube is 1}) \end{aligned} \quad (9)$$

Note. There is a bug with this process. From our reasoning, it follows that the volume of $V(\mathbf{e}_2, \mathbf{e}_1)$ is -1. It seems like this ordering of edges is as valid a way to describe the unit square, so why is the area negative? We can fix this “bug” by claiming that it is, in fact, a feature: we haven’t just computed the area/volume, we have computed the *oriented volume* – the determinant gives you more than the volume alone. If you want the volume you need only take the absolute value of the oriented volume.

We write the final formula down for reference but now with the caveat that the determinant of a matrix is not a positive stretching factor – the amount that the volume of the unit cube is stretched under A , but

²More generally, permutations on n objects are bijections (one-to-one *and* onto mappings) from the set of integers $\{1, 2, \dots, n\}$ to itself.

³ S_n is the symmetric group of n objects. It consists of all permutations that can be performed on n objects.

⁴The mapping $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a group homomorphism; $\text{sgn}(\sigma) = -1$ if k is odd and $\text{sgn}(\sigma) = +1$ if k is even, where k is the number of transpositions in any factorization of σ (this parity is well-defined).

a *signed* stretching factor. Again, to find the magnitude of the stretching, take the absolute value of the determinant.

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \cdots a_{\sigma(n),n} \quad (10)$$

Existence. Our derivation shows that *if* a function with the three properties exists, it must be given by (10) – a *uniqueness* argument. To know the three properties are not secretly contradictory, one must also check that the formula (10) actually satisfies them. Multi-linearity is immediate: each product contains exactly one factor from each column. The value on the canonical cube is 1: only the identity permutation survives when $A = I$. Degeneracy takes one idea: if two columns are equal, pair each permutation σ with $\sigma \circ \tau$, where τ is the transposition swapping the two equal column indices; the paired products are equal (the swapped factors come from identical columns) while their signs are opposite, so the sum cancels in pairs to 0.

1.1 Determinant of A^T

Notice that the formula (10) can also be written as:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(T(\sigma)) a_{T(\sigma)(1),1} a_{T(\sigma)(2),2} \cdots a_{T(\sigma)(n),n} \quad (11)$$

Here, T is any bijection from S_n to itself. Why? Because the sum in (10) is the sum over all elements of S_n . The mapping T just rearranges that sum. One such choice for T is the mapping that sends σ to its inverse, σ^{-1} . In this case, (11) becomes:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a_{\sigma^{-1}(1),1} a_{\sigma^{-1}(2),2} \cdots a_{\sigma^{-1}(n),n} \quad (12)$$

Reorder the factors in each product (multiplication is commutative) and re-index by $i = \sigma^{-1}(k)$, so that $k = \sigma(i)$:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a_{1,\sigma(1)} a_{2,\sigma(2)} \cdots a_{n,\sigma(n)} \quad (13)$$

Letting $a'_{i,j}$ denote the (i,j) entry of A^T (so $a'_{i,j} = a_{j,i}$), we have $a_{i,\sigma(i)} = a'_{\sigma(i),i}$, hence:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma^{-1}) a'_{\sigma(1),1} a'_{\sigma(2),2} \cdots a'_{\sigma(n),n} \quad (14)$$

Since $\text{sgn} : S_n \rightarrow \{\pm 1\}$ is a group homomorphism, $\text{sgn}(\sigma) \cdot \text{sgn}(\sigma^{-1}) = \text{sgn}(\epsilon) = 1$,⁵ so $\text{sgn}(\sigma^{-1}) = \text{sgn}(\sigma)$ (since ± 1 are self-inverse). With this, (14) may be written:

$$|A| = \sum_{\sigma \in S_n} \text{sgn}(\sigma) a'_{\sigma(1),1} a'_{\sigma(2),2} \cdots a'_{\sigma(n),n} \quad (15)$$

But this is, by definition, nothing more than the determinant of A^T . From this we conclude that $|A| = |A^T|$.

⁵ ϵ denotes the identity in S_n .

1.2 Determinant of a Product

The derivation of (10) proved more than a formula; it proved a *uniqueness* statement. Any function W of n vector arguments that is multi-linear and has the Degenerate Property – it vanishes whenever two arguments are equal; the standard name for such a function is *alternating* – satisfies

$$W(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = \left(\sum_{\sigma \in S_n} \text{sgn}(\sigma) a_{\sigma(1),1} a_{\sigma(2),2} \dots a_{\sigma(n),n} \right) W(\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_n) \quad (16)$$

where, as before, the $a_{i,j}$ are the components of the $\mathbf{x}_j$ in the standard basis – the computation is the same as the one leading to (10), only without using the third property to set $W(\mathbf{e}_1, \dots, \mathbf{e}_n) = 1$. In words: such a W equals its value on the canonical cube times the determinant of the matrix whose columns are its arguments.

This buys us the product rule for determinants almost for free. Fix a matrix A and define

$$W(\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n) = V(A\mathbf{x}_1, A\mathbf{x}_2, \dots, A\mathbf{x}_n).$$

Then W is multi-linear (because A is linear and V is multi-linear) and alternating (equal slots have equal images under A , on which V vanishes). Apply (16) to the columns $\mathbf{b}_1, \dots, \mathbf{b}_n$ of a matrix B , noting that $A\mathbf{b}_j$ is the j^{th} column of AB and that $W(\mathbf{e}_1, \dots, \mathbf{e}_n) = V(A\mathbf{e}_1, \dots, A\mathbf{e}_n) = |A|$:

$$|AB| = W(\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n) = |A| |B| \quad (17)$$

This is the *Multiplicative Property* of the determinant; we will use it in the next section.

It also pays an immediate dividend, promised in the opening: if A^{-1} exists, then $|A| |A^{-1}| = |AA^{-1}| = |I| = 1$, so $|A| \neq 0$. Read contrapositively: a matrix with determinant 0 – one that flattens the unit cube to zero volume – cannot be inverted. (The converse – that $|A| \neq 0$ guarantees A^{-1} exists – is also true, but requires machinery we do not build here; we flag it as an honest debt.)

2 Derivative of the Determinant

Here is a question to hold while we build the next tool. Suppose n vectors in R^n are being stirred by a flow – each evolving according to $\dot{\mathbf{x}} = A\mathbf{x}$. The parallelepiped they span is carried along; *how does its volume evolve in time?* The volume at time t is a determinant, so to track it we will need the derivative of the determinant. That is what this section computes; the next section collects the payoff.

The determinant is a function from the space of matrices to R ; that is, $\det : R^{n \times n} \rightarrow R$. Therefore, its derivative at a particular “point” A , L_A , should satisfy:⁶

$$|A + H| = |A| + L_A(H) + o(H) \quad (18)$$

Here $\lim_{H \rightarrow 0} \frac{|o(H)|}{\|H\|} = 0$, where $\|H\|$ is any matrix norm and L_A is a linear mapping from $R^{n \times n}$ to R .

⁶Recall that for a function $f : R \rightarrow R$, the derivative at a point x is a linear function, $Df_x : R \rightarrow R$, such that $f(x+h) = f(x) + Df_x(h) + o(h)$.

To start, let's try to find the derivative for an especially simple “point”, the identity matrix, I .

$$|I + H| = \prod_{i=1}^n (1 + H_{i,i}) + o(H) \quad (19)$$

Only the identity permutation contributes to the linear-in- H part of $|I + H|$. To see this, take any non-identity $\sigma \in S_n$. Its non-fixed points form cycles of length ≥ 2 , so there are at least two indices i with $\sigma(i) \neq i$. For each such i , the entry $(I + H)_{\sigma(i),i} = H_{\sigma(i),i}$ is off-diagonal (no 1 contribution). Hence the product for σ contains at least two off-diagonal H entries, so it is $O(\|H\|^2) = o(\|H\|)$. This product can be further decomposed into constant, linear-in- H , and $o(H)$ terms:

$$\begin{aligned} |I + H| &= 1 + \sum_{i=1}^n H_{i,i} + o(H) \\ &= |I| + \text{tr}(H) + o(H) \end{aligned} \quad (20)$$

So, the linear function acting on H to produce a number is the trace operator.

Now, let's do the same computations for a more general “point”, an $n \times n$ matrix, A , with the stipulation that A is invertible.

$$\begin{aligned} |A + H| &= |A(I + A^{-1}H)| \\ &= |A| |I + A^{-1}H| \quad (\text{Multiplicative Property (17)}) \\ &= |A| (|I| + \text{tr}(A^{-1}H) + o(H)) \quad (\text{From (20)}) \\ &= |A| + |A|\text{tr}(A^{-1}H) + o(H) \end{aligned} \quad (21)$$

Therefore, the derivative of the determinant at a non-singular “point”, A , is the linear function $L_A : R^{n \times n} \rightarrow R$ defined by:

$$L_A(B) = |A|\text{tr}(A^{-1}B) \quad (22)$$

This is known as *Jacobi's formula* for the derivative of the determinant.⁷

Notice the strategy just used: we computed the derivative where it is easy – at the identity – and then let the Multiplicative Property *transport* the answer to every invertible point, by rewriting $A + H$ as $A(I + A^{-1}H)$. A question about a general point became a question about the easiest point; this is the “change the representation” move, and it is worth remembering.

It is also worth pausing to see which game this section has been playing. “Derivative = best linear approximation” is a definition the reader already owns for functions of one variable; nothing about it changed except the size of the board:

⁷The invertibility assumption on A can be removed. The general form of Jacobi's formula is $L_A(B) = \text{tr}(\text{adj}(A)B)$, where $\text{adj}(A)$ is the adjugate (transpose of the cofactor matrix). This expression is polynomial in the entries of A and so is defined for all A , singular or not. When A is invertible, $\text{adj}(A) = |A|A^{-1}$ and the two forms agree; when A is singular, the adjugate form is the right one.

Function	Derivative at a point	Acting on an increment
$f : R \rightarrow R$	a number, $f'(x)$	$f'(x) h$
$f : R^n \rightarrow R$	a vector, $\nabla f(\mathbf{x})$	$\nabla f(\mathbf{x}) \cdot \mathbf{h}$
$\det : R^{n \times n} \rightarrow R$	a linear form, L_A	$ A \operatorname{tr}(A^{-1}H)$

Same definition, bigger board; the only thing that grows is what “a linear function of the increment” can look like.

Exercise. Check that L_A is a linear function.

3 Applications: Liouville’s Theorem

Suppose that a dynamical system is deforming n vectors in R^n over time by the following linear *vector differential equations*:

$$\frac{d\mathbf{x}_i(t)}{dt} = A\mathbf{x}_i(t) \quad i \in [1, n] \quad (23)$$

$$\mathbf{x}_i(0) = \mathbf{x}_{0,i} \quad i \in [1, n] \quad (24)$$

Question. How is the volume of the parallelepiped formed from the vectors $\{\mathbf{x}_{0,i}\}_{i=1}^n$ evolving in time?

To start, we could collect these vector equations into one linear *matrix differential equation*:

$$\frac{dX(t)}{dt} = AX(t) \quad (25)$$

$$X(0) = X_0 \quad (26)$$

Here, the n vectors form the columns of the matrix $X(0)$. The columns of $X(t)$ represent the evolution of the vectors, $\{\mathbf{x}_i\}_{i=1}^n$, in time. The determinant of $X(t)$, $|X(t)|$, represents the evolution of the volume of the parallelepiped represented by these vectors.

To answer the question posed, we need to track $|X(t)|$. First note that the solution of (25) is $X(t) = e^{tA}X_0$, where e^{tA} is the *matrix exponential*.⁸ Since e^{tA} is invertible for *every* t – its inverse is e^{-tA} – the matrix $X(t)$ is invertible for all time whenever X_0 is. (And if $|X_0| = 0$, then by the Multiplicative Property $|X(t)| = |e^{tA}||X_0| = 0$ for all t , so the formula we derive below holds trivially.) Assume then that $|X_0| \neq 0$. We find a differential equation that $|X(t)|$ satisfies and then solve.

$$\frac{d|X(t)|}{dt} = |X(t)| \operatorname{tr} \left(X(t)^{-1} \frac{dX(t)}{dt} \right) \quad (\text{by (22) and the chain rule})$$

$$\frac{d|X(t)|}{dt} = |X(t)| \operatorname{tr} (X(t)^{-1} AX(t)) \quad (\text{by (25)})$$

$$\frac{d|X(t)|}{dt} = |X(t)| \operatorname{tr}(A) \quad (\text{by the property of trace: } \operatorname{tr}(ABC) = \operatorname{tr}(CAB))$$

⁸ $e^{tA} \equiv \sum_{k=0}^{\infty} \frac{t^k A^k}{k!}$; one checks term by term that $\frac{d}{dt} e^{tA} = A e^{tA}$ and that $e^{tA} e^{-tA} = I$.

Therefore, $|X(t)|$ satisfies the following linear *scalar differential equation*:

$$\frac{d|X(t)|}{dt} = \text{tr}(A) |X(t)| \quad (27)$$

$$|X(0)| = |X_0| \quad (28)$$

This is much easier to solve; the solution is $|X(t)| = |X_0| e^{\text{tr}(A)t}$. This is *Liouville's formula*; it is a special case of Liouville's theorem on phase-space volume preservation in dynamical systems. (The chain-rule step above used $X(t)^{-1}$, which is legitimate because we established up front that $X(t)$ remains invertible for all time.) Note the move: a matrix differential equation we could not track directly became a scalar differential equation solvable by first-year calculus – change the representation, again.

Question. What kind of systems (defined by A) leave the volumes unchanged?

From the differential equation $\frac{d|X(t)|}{dt} = \text{tr}(A) |X(t)|$ and its solution $|X(t)| = |X_0| e^{\text{tr}(A)t}$, the volume is preserved exactly when $\text{tr}(A) = 0$. An important sub-class is anti-symmetric matrices: when $A^T = -A$, then $a_{ii} = -a_{ii}$ forces $a_{ii} = 0$ for every i , so $\text{tr}(A) = 0$ and volumes are preserved.

Anti-symmetric A in fact gives a *stronger* conclusion than mere volume preservation. From $A^T = -A$ we have $\langle A\mathbf{x}, \mathbf{x} \rangle = \langle \mathbf{x}, A^T \mathbf{x} \rangle = -\langle \mathbf{x}, A\mathbf{x} \rangle = -\langle A\mathbf{x}, \mathbf{x} \rangle$, so $\langle A\mathbf{x}, \mathbf{x} \rangle = 0$. Applied to the trajectory:

$$\left\langle \mathbf{x}(t), \frac{d\mathbf{x}(t)}{dt} \right\rangle = \langle \mathbf{x}(t), A\mathbf{x}(t) \rangle = 0,$$

which gives $\frac{d}{dt} \|\mathbf{x}(t)\|^2 = 2 \langle \mathbf{x}, \dot{\mathbf{x}} \rangle = 0$: the *individual length* of each evolving vector is preserved, not just the volume of the parallelepiped they span. In fact, the same computation applied to two trajectories gives

$$\frac{d}{dt} \langle \mathbf{x}_i(t), \mathbf{x}_j(t) \rangle = \langle A\mathbf{x}_i, \mathbf{x}_j \rangle + \langle \mathbf{x}_i, A\mathbf{x}_j \rangle = \langle A\mathbf{x}_i, \mathbf{x}_j \rangle - \langle A\mathbf{x}_i, \mathbf{x}_j \rangle = 0,$$

so *all* pairwise inner products – lengths *and* angles – are preserved by the flow.

Exercise. Suppose $X_0 = I$. In this case, what kind of matrix is $X(t)$?

Curtain call. We can now answer the opening question in its own words. The determinant was never a formula to memorize; it was the answer to “by what factor does A change volume?” Three properties that any oriented volume should have *forced* the formula (10) – the sign was the price the algebra charged, and what it bought was orientation. The same three properties, through the uniqueness statement (16), delivered $|AB| = |A| |B|$, which told us that a matrix that flattens volume can never be inverted, and which transported the derivative from the identity to every invertible matrix: $L_A(H) = |A| \text{tr}(A^{-1}H)$. And that trace formula answered a question about motion: under the flow $\dot{\mathbf{x}} = A\mathbf{x}$, volume grows like $e^{\text{tr}(A)t}$ – so volume is conserved exactly when $\text{tr}(A) = 0$.

Companion papers. The step “ $A\mathbf{e}_j$ is the j^{th} column of A ,” on which the whole derivation stands, is itself derived – forced by the demand that the scalar solution of $ax = b$ generalize to systems – in “Origins

of Matrix/Vector and Matrix/Matrix Multiplication.” The volume-stretch question survives even when the matrix is not square: in “Singular Transformations of Probability Density Functions,” a matrix $A : R^n \rightarrow R^m$ stretches n -dimensional volume by the *Gram factor* $\sqrt{|A^T A|}$; when A is square, the two theorems proved here collapse that factor to the present answer, $\sqrt{|A^T A|} = \sqrt{|A^T| |A|} = \sqrt{|A|^2} = ||A||$ – the transpose rule and the product rule working together in one line. The ladder of section 2 – derivative as best linear approximation, with the board growing from slopes to gradients to a trace form – is the backbone of “What is a Derivative” and of the differential-approximation chapter of the author’s calculus book, where $|\det Df|$ reappears as the local volume ledger of a nonlinear map. And lengths and angles – the quantities the anti-symmetric flows above preserve – are exactly what the inner product measures; where *that* object comes from is the subject of “Where does the Dot Product Come From?”

Further reading. The axiomatic treatment of the determinant given here follows the classical route; see K. Hoffman and R. Kunze, *Linear Algebra* (Chapter 5), or M. Artin, *Algebra*. For a spirited contrarian view of the whole subject, see S. Axler’s essay “Down with Determinants!” (*American Mathematical Monthly*, 1995).